FLOW A: Intelligent Multi-Agent Portfolio Generation
Investor Onboarding (Risk Profile, Goal, Horizon, Capital)
Manager Node (Task Decomposition & Context Assembly)
Suitability & Feasibility (SEBI Risk Alignment & Lot Feasibility)
Fail
Pass
Macro Analyst Node (RBI Policy, Repo Rate, CPI Inflation)
Market Analyst Node (Nifty 50 Regime, 50/200 DMA)
Fundamental Analyst Node (P/E, ROE, D/E, Factor Scoring)
News & Filings Analyst Node (30-Day Recency Filter, Filings)
Lead Strategist Node (Tactical Weights & Regime Thesis)
Independent Critic Node (Mandate, Caps, Liquidity Audit)
Reject / Exhausted
Approved
Rework (<=2)
Deterministic SLSQP Optimizer (Markowitz Quadratic Utility)
Strategist Bounded Rework Node
Critic Final Validation (Post-Optimizer Constraint Check)
Violation
Approved
Persistence Node (Aborted Status & Actionable Notice)
Fiduciary Moderator Node (Two-Level Institutional Explanations)
ACID PostgreSQL Persistence (Commit Portfolio & Audit Traces)
Client Presentation & Execution Review
FLOW E: Conversational AI & Fiduciary 'Stock-Why'
User Natural Language Query ('Why is HDFCBANK in my portfolio?')
NLP Profile & Intent Extractor (Extract Ticker, Query Type, Risk Profile)
Stock Why Service Aggregator (Retrieve Strategy Thesis, Fundamentals, Macro Data)
Two-Level Explanation Engine (Level 1: Plain English | Level 2: Institutional Quant Factors)
Fiduciary Compliance Validation (SEBI RIA Safe Harbor Disclaimer & Fact Check)
Interactive Chat Response (Citations, Financial Ratios, Regime Justification)
FLOW D: Demat Import & Portfolio Health Diagnosis
Existing Holdings Submission (Manual Demat Entry or Structured CAS JSON)
Security Master Reconciliation (Resolve ISINs across 2,639 NSE/BSE Universe)
Demat Diagnosis Service Engine (Analyze Health, Overlap, Factor Tilts)
Valuation & Factor Stress Testing (Weighted P/E, P/B, Debt/Equity, Beta)
Compute Concentration Risk (Herfindahl-Hirschman Index HHI & Sector Skew)
Generate Diagnostic Report & Rebalance Recommendations (Specific Trims, High-Conviction Replacements)
FLOW B: Automated Portfolio Drift Monitoring
Background Scheduler (Every 300s, Market Hours 09:15-15:30 IST)
Fetch Live NSE/BSE Quotes (Market Data Adapter / Yahoo Finance)
Update Market Prices & Valuation (Current Portfolio NAV & Position Values)
Compute Drift Metrics (Asset Drift & Sector Allocation Drift)
Drift Threshold Breach? (Stock >= 5.0% OR Sector >= 8.0%)
No Breach
Breach Detected
Healthy Status Logged (Sleep until next cycle)
Trigger Rebalancing Alert (In-App Notification & Audit Log)
FLOW C: Unified Portfolio Rebalancing Engine
Rebalance Triggered (Drift Breach OR User Rebalance Request)
Retrieve Target Allocation (Active Strategy Definition & Cap Rules)
Compute Required Delta (Target Capital Value - Current Market Value)
Whole-Share Integer Allocation (Math Floor Quantity & Integer Rounding)
Cash Conservation Invariant (Invested + Cash == Total NAV, Residual Cash >= 0)
Statutory Indian Transaction Costs (STT + GST + Stamp Duty + Exchange/SEBI)
Generate Trade Preview (BUY, SELL, HOLD Orders with Rationale)
User Discretionary Review (Non-Custodial Advisory Gate)
Approve
Create Portfolio Version (N+1) (ACID DB Commit & Order Instructions)